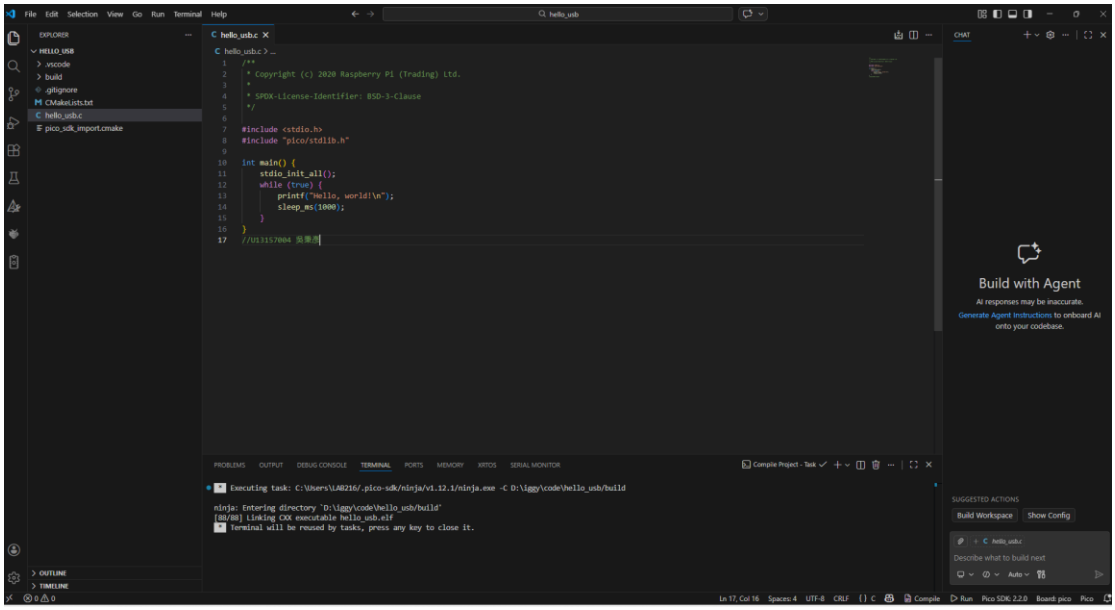
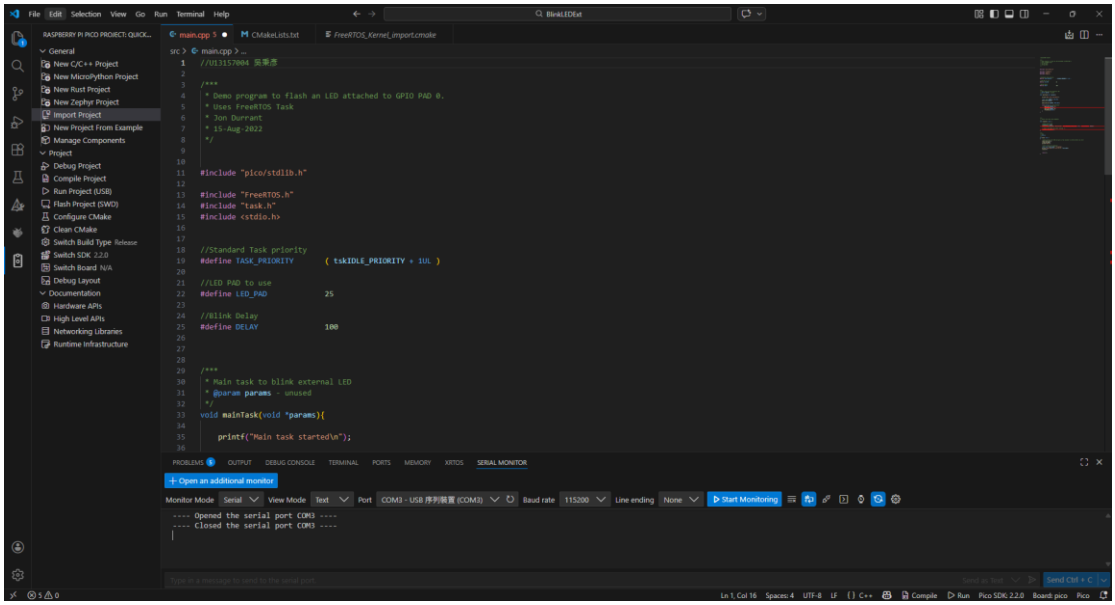
HW3

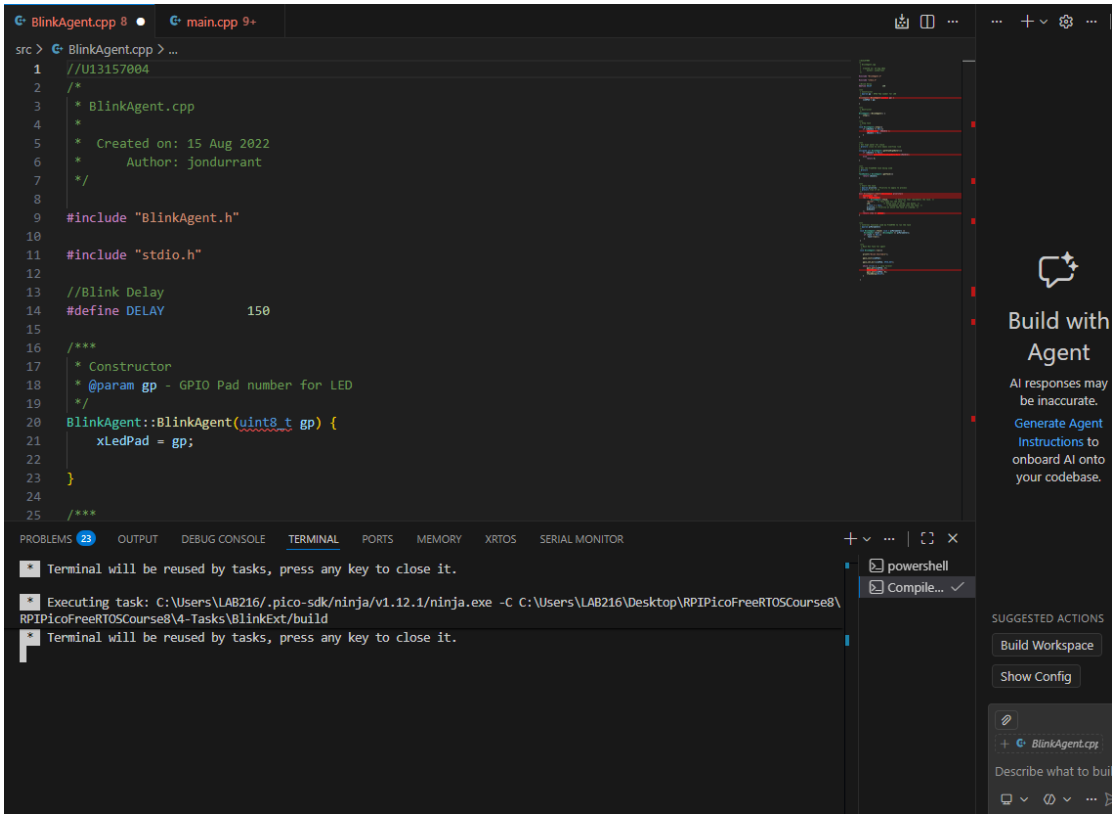
Ch0


hello_usb.c
/**
* Copyright (c) 2020 Raspberry Pi (Trading) Ltd.
*
* SPDX-License-Identifier: BSD-3-Clause
*/
#include <stdio.h>
#include "pico/stdlib.h"
int main() {
stdio_init_all();
while (true) {
printf("Hello, world!\n");
sleep_ms(1000);
}
}
//U13157004 吳秉恩
Build with Agent
Executing task: C:\Users\LA0216/.pico-sdk/ninja/v1.12.1/ninja.exe -C D:\iggy\code\hello_usb\build
ninja: Entering directory `D:\iggy\code\hello_usb\build'
[8/8] Linking CXX executable hello_usb.elf
Terminal will be reused by tasks, press any key to close it.

Ch2


main.cpp
//U13157004 吳秉恩
/***
* Demo program to flash an LED attached to GPIO PAD 0.
* Uses FreeRTOS Task
* Jon Durrant
* 15-Aug-2022
*/
#include "pico/stdlib.h"
#include "FreeRTOS.h"
#include "task.h"
#include <stdio.h>
//Standard Task priority
#define TASK_PRIORITY (tskIDLE_PRIORITY + 1UL)
//LED PAD to use
#define LED_PAD 25
//Blink Delay
#define DELAY 100
/***
* Main task to blink external LED
* @param params - unused
*/
void mainTask(void *params){
printf("Main task started\n");
---- Opened the serial port COM3 ----
---- Closed the serial port COM3 ----

Ch4



比較表

項目	Ch0 – SDK Example	Ch2 – BlinkLEDExt	Ch4 – BlinkExt (FreeRTOS)
開發方式	官方 SDK 範例	自建專案	使用 FreeRTOS 系統
程式結構	單一 `while(true)` 迴圈	單一 `while(true)` 迴圈	使用 Task 管理多任務
延遲方式	`sleep_ms()`	`sleep_ms()`	`vTaskDelay()`
RTOS 支援	否	否	是
多工支援	否	否	是

README

比較 SDK example、BlinkExt、Blink 程式，了解不同編譯過程和實現方式。